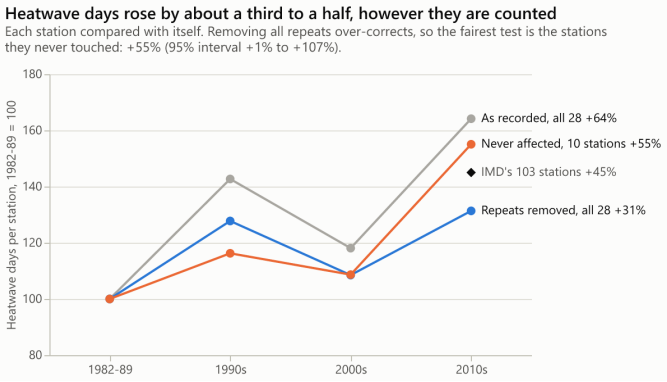
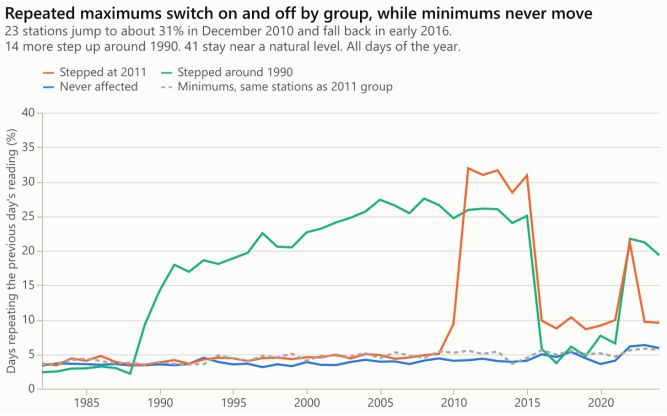
India's Heat Record: what 1.07 billion weather rows can and cannot show

Ritik Yadav, November 2026. 1.07B station-days from NOAA GHCN-Daily, built from 2.68B measurements, analysed in ClickHouse. Results describe the stations named here, not all of India.

Bottom line. Heatwave days did rise at these stations, by about a third to a half from the 1980s to the 2010s, in line with IMD's own count. But the public record carries an undocumented reporting artefact that switches on region by region and inflates the count where it applies. Accounting for it changes the answers: nights are not outpacing days, summer's first 40°C has not clearly moved, and rain after 1972 cannot be compared with rain before it.





1. Heatwave days rose by about a third to a half

At 28 stations compared with themselves, heatwave days per hot season rose 64% from 1982-89 to the 2010s as recorded. Removing every repeated reading gives 31% (95% interval -6% to 70%), but that also drops genuine hot days. The fairest test is the 10 stations the problem never touched: up 55% (interval 1% to 107%), though 4 of them fell. IMD's own count at 103 stations rose 45%.

2. A reporting artefact switches on by region

A daily maximum identical to the previous day's normally covers about 4% of days. At 23 stations it jumps from 14% in November 2010 to 31% a month later, holds for five years, then falls to 8% by March 2016. Minimums at the same stations never move, and no station jumps by 5 to 15 points: it is all or nothing. 22 of the 23 stations in the area of IMD's Chennai centre are in this group, and none of Maharashtra's 7. Another 14 in the north-west step up around 1990. These repeats sit on local peaks 6 times as often as on troughs, against 1.3 at clean stations, which is what copying a day's peak into the next day would do. The cause is inferred, not documented. At the stations that stepped at 2011, heatwave days rise 46% as recorded, and 6% without repeats.

3. Nights are not outpacing days

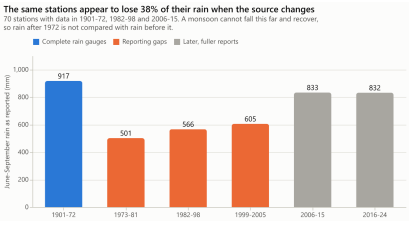
From 1982 to 2024 the median station warmed +0.28°C a decade by day and +0.21°C by night, and nights led at 11 of 31 stations. Much of that gap comes from affected stations: at the 13 clean ones the two are level (+0.28 and +0.27°C), with nights ahead at 8. IMD's warm nights, which count only on days reaching 40°C, became more common but remain rare.

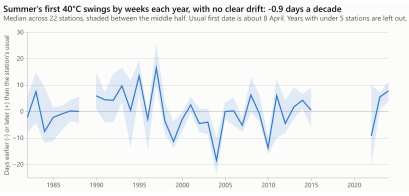
4. Summer's start and monsoon extremes show no clear shift

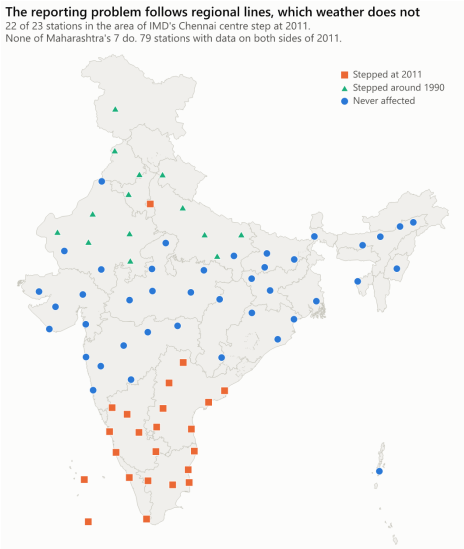
The first 40°C moved -0.9 days a decade across 22 stations (interval -2.5 to +1.2), while years swing by 35 days. Across 1,714 rain gauges from 1901 to 1970, heavy days (100 mm or more) went from 0.79 to 0.84 a season. Coverage then falls from 2,425 stations to 84 by 1973, and the same 70 stations appear to lose 38% of their rain before recovering, so no trend is drawn across that break. Within 2006-2024, 76 stations show no clear change either (+0.04 heavy days a decade, interval -0.07 to +0.15).

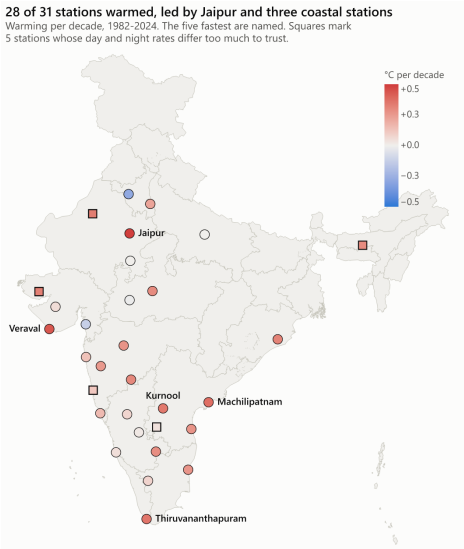
5. India's stations warmed more slowly than most countries'

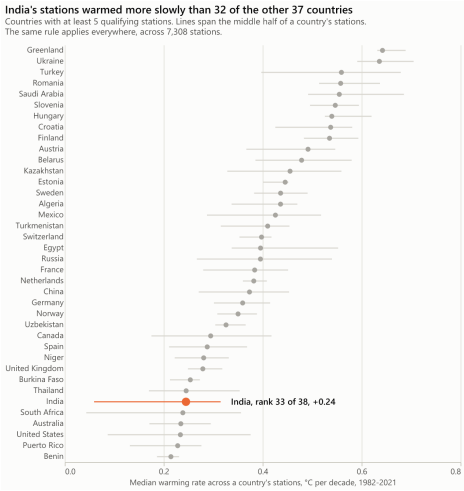
28 of 31 stations warmed, with a median of +0.25°C a decade. Under one rule for every country, 1982-2021, India ranks 33 of 38 at +0.24°C, against a median country of +0.40°C, or 28 of 38 using clean stations alone. Slower warming does not mean little heat: the 2024 hot season averaged 8.4 heatwave days at the 10 stations with complete data, against 4.9 in their 1991-2020 average, with repeats removed.

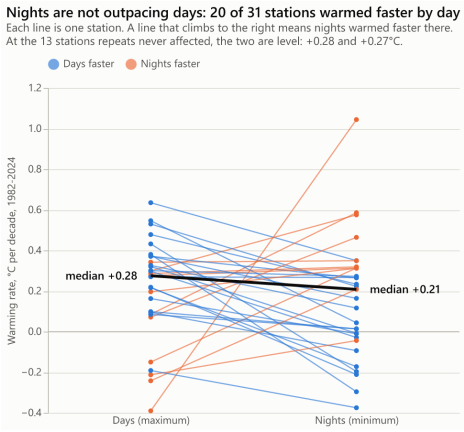












How it was built

ClickHouse holds all 1,074,089,837 rows in 1.88 GB, smaller than the 2.28 GB Parquet they came from, using per-column codecs, a table partitioned by decade and sorted by station and date, and rollups kept current by materialized views. The rollup returned the same answer as a full scan on all 7,308 stations. Times below are real, and include wherever DuckDB wins.

Median warm seconds, 4 threads, 3 GB	ClickHouse raw	ClickHouse rollup	DuckDB on Parquet
India, every station-year	1.08	0.12	0.59
One station's whole history	0.08	0.07	0.12
Every country by decade, full scan	15	0.13	12
Warming rate, every station: first query	32	7.33	6.74
Warming rate, every station: rewritten query	20	2.34	6.58

Rise in heatwave days, 1982-89 to the 2010s, if one choice changes (all in the README)	Stations	As recorded	Repeats removed
This project's rule	28	+64%	+31%
Normals from 1981-2010, not 1991-2020	28	+72%	+36%
The plan's simpler rule: 40 C and 4.5 C above normal, everywhere	28	+79%	+38%
Only stations repeats never affected	10	+55%	+59%
Leaving out stations that stepped at 2011	14	+77%	+50%

What this cannot tell you

These are stations, not India. The file names 3,806 Indian stations, but only 28 pass the bar for heatwave trends, none in the hills, and few have complete seasons after 2015. **One source.** Every Indian maximum through 2024 comes from a NOAA source its readme says to use with caution, and GHCN-Daily does not adjust for station moves. 5 stations show day and night rates too far apart to trust. **Not IMD's declarations.** IMD declares heatwaves by sub-division, and has since added percentile thresholds. This applies IMD's published station rule to every year alike. **Published figures differ in kind.** IMD's 536 heatwave days in summer 2024 adds up sub-divisions, so it cannot be compared with counts per station here. Temperature stops in 2024, because NOAA changed these stations' source in 2025.

Data: Menne et al. 2012, GHCN-Daily, via ClickHouse and NOAA. Full checks: docs/issue_log.csv (15 checks). Corrections to the original plan: docs/plan_corrections.md. Sources: docs/sources.md.